Case Study: ChatGPT - Measuring and Growing Our Core Business

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Key performance indicators and Guiding Stars

Morning Gym KPIs

Metric Category

Metric

Rationale

Expected Impact if Optimized

Relevant Benchmarks

Engagement

Task Completion Rate

Tracks the percentage of prompts where users feel their task was successfully completed.

A 5% increase could improve satisfaction, driving a 10% uplift in retention and repeat usage.

Target: ?80% completion satisfaction rate.

Daily Active Users (DAUs) by Tier

Monitors activity segmentation (free, Plus, Pro, Team and enterprise users).

Growing DAUs by 10% MoM in paid tiers could increase ARR by $20M for every additional 1M users.

SaaS target DAU growth: 5-10% MoM.

Multi-Session Daily Usage

Tracks users who engage with ChatGPT more than once per day.

Boosting multi-session usage by 10% could reflect deeper utility, raising paid tier upgrades by 5-7%.

Benchmark: ?30% of DAUs using multiple sessions.

Monetization

Subscription Retention Rate

Measures retention across different subscription tiers

A 5% increase in annual retention could add $50M ARR, assuming steady conversion rates.

Target SaaS retention: ?85% annually.

Cost of Acquisition per Subscriber

Calculates how much is spent on acquiring each new paying user.

Reducing CAC unlocks higher marketing efficiency and increases ROI on campaigns.

Benchmark CAC: ?2x ARPU.

Average Revenue Per User (ARPU)

Captures earnings across free and paid tiers.

Improving ARPU by 10% across all tiers could drive sustainable revenue growth of $50-$100M annually.

SaaS ARPU varies widely, $10-$25 for similar apps.

Operational

Feature-Level NPS

Assesses user sentiment for individual features (e.g., summarization, image generation).

Enhancing feature NPS by 10 points correlates to a 15% increase in feature adoption and upsell conversions.

NPS benchmark: ?40 for key SaaS features.

Perplexity Score

Measures how well the model predicts a sample; lower scores indicate better performance.

Reducing perplexity by 10% improves response relevance, boosting completion rates by 15%.

A perplexity score below 20 is considered excellent for language models.

Cost per Query

Tracks efficiency in delivering AI-powered responses.

Cutting query costs by 15% yields an annual margin improvement of $100M at scale for large query volumes (~10B/month).

Estimated cost: $0.002-$0.03 per query.

Guiding Stars

Retention-Driven NSM - 7-Day Retention Rate for Core Users

Definition: The percentage of new users who return within 7 days of their first session.

Rationale: High retention indicates ChatGPT is becoming indispensable. This correlates to habit formation and better long-term monetization.

Quantitative Impact: Improving retention by 10% could add 20% to MAUs and boost free-to-paid rate by ~15%

Monetization-Driven NSM - Paid Subscriber LTV-to-CAC Ratio

Definition: Calculates lifetime value (LTV) of paying users compared to their acquisition cost (CAC).

Rationale: A rising ratio signals profitable growth and better marketing ROI.

Quantitative Impact: Improving the LTV-to-CAC ratio from 2.5x to 4x could unlock larger marketing budgets.

Engagement-Driven NSM - Monthly Prompts Per User

Definition: Average number of user queries per month.

Rationale: Queries signal depth of use and reliance. Growth in this metric often translates into better monetization, as active users are likely to subscribe to premium features.

Quantitative Impact: Increasing prompts/user by 15% could lift subscription revenues by ~10%.